



Stronger together: a cross-SIG perspective on improving drug development

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Introduction

Pharmacometric modeling and simulation (M&S) plays an increasingly important role in drug development. In order to bring new treatments to patients safely and quickly, today's leaders need to leverage biology, pharmacology, and computational technology with M&S to make the best decisions at all stages of development. By combining these different sources of knowledge, M&S improves decision making throughout the lifecycle of a drug.

The International Society of Pharmacometrics (ISoP), a non-profit organization, brings together global leaders in pharmacometrics (PMx) who are committed to advancing and promoting this field [1]. Its primary objective is to support its members and the broader community to improve

regulatory and therapeutic decision-making processes through the application of scientific principles. The special interest groups (SIGs) within ISoP, along with the working groups within these SIGs, have a pivotal role in expanding model-informed drug development (MIDD) principles and methodology to “integrate knowledge across the development program and compounds and biology” [2]. Each of the SIGs have their own perspectives and ideas, but they share the goal of improving drug development and patient outcomes. Our individual accomplishments are only possible, or impactful, because of our collective efforts and advancements in the field of pharmacometrics and systems pharmacology, which are ultimately how we improve patient's lives.

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In this paper, we have summarized the missions and goals of the ISoP SIGs and discuss how each SIG integrates pharmacometric (PMx) approaches into MIDD to facilitate drug development. A view of the future roles and activities of the ISoP and each of the SIGs is presented, including how we can train the next generation of pharmacometricians and leaders, how we can collaborate across disciplines, and the impact of innovations in methodology and resources. We believe the future of pharmacometrics in drug development is promising.

SIG missions

The field of pharmacometrics has built on the innovations developed in other fields, such as engineering, mathematics, statistics, physics, and medicine. The five current SIGs within ISoP reflect these foundations as we continue to build upon them. As of July 2024, the SIGs are: Quantitative Systems Pharmacology (QSP); Mathematical and Computational Sciences (MCS); Statistics and Pharmacometrics (SxP); Clinical Pharmacometrics (Clin PMx); and Pharmacometrics Data Programming (PMxP). Table 1 describes the roles and remit of each SIG and working groups within the SIGs. Although driven by different sciences and approaches, the five SIGs have shared common goals (Fig. 1).

Quantitative systems pharmacology (QSP) SIG

QSP models are mechanistic mathematical models that may include the (patho)physiology of interest, mechanistic links between target modulation and key endpoints and/or biomarkers, the overall dynamics of the system, the population variability, and pharmacologic and other intervention(s). By assimilating all available knowledge and data, QSP modeling can inform decision-making during the lifecycle of a drug beginning at the preclinical stage through post-approval. The goal of the QSP SIG is to advance the development and utilization of safe and efficacious medicines through the application of QSP modeling.

Mathematical and computational sciences (MCS) SIG

The primary mission of the MCS SIG is to promote the development and use of mathematical and computational techniques in pharmacometrics [3]. With the rapid speed at which computational resources are advancing, novel approaches to the theory, methodology, and computational tools for pharmacometrics and systems pharmacology modeling need to be developed and expanded. The MCS SIG works closely with the Society for Industrial and Applied Mathematics Life Sciences Activity Group and with the Society for Mathematical Biology. Close ties ensure that

Table 1 The roles and remit of current ISoP SIGs and working groups within the SIGs

Special interest group	Quantitative systems pharmacology (QSP)	Mathematical and computational sciences (MCS)	Statistics and pharmacometrics (SxP)	Clinical pharmacometrics (Clin PMx)	Pharmacometrics data programming (PMxP)
Year Founding	2014	2017	2016	2017	2024
Stages of development	200+ members with 11 steering committee and leadership team	70+ members with 12 steering committee members, including 3 elected officers	200+ general members with 12 elected steering committee members and 4 leadership team members; Special emphasis (sub-committees) in communication and education	5 leadership team members, 8 steering committee members	~50 members with 12 steering committee members
Working Groups	Neurosciences; Inflammation and Immunology; Integrating QSP and Machine learning.	Modeling Delays in PKPD; Optimal Control; Sensitivity Analysis/Uncertainty Quantification; AI/Machine learning	Model-based meta-analysis; AI/ML	Outreach & Engagement task force and Education task force	PopPK and PK/PD dataset; Data visualization, exploration and source/PMX data check; QSP Programming; Implementation of CDISC PopPK
External Affiliations and/or Partner Associations			American Statistical Association	American College of Clinical Pharmacology	
Websites	https://sites.google.com/view/isopqpsig/	https://sites.google.com/view/mcssig/	https://sxpsig.github.io/	https://www.accp1.org/Members/About/ACCP1/0About/ISoP/ACCP_Special_Interest_Group.aspx	https://www.isop.org/special-interest-groups/pmxsig



Fig. 1 Common goals of ISoP SIGs. The flower diagram shows that all of the ISoP SIGs share common goals in communication, sciences, leadership, collaboration, education, and supporting regulatory agencies. ISoP, international society of pharmacometrics; SIG, special interest group; ACoP, American conference on pharmacometrics

important problems and applications are shared with researchers outside of ISoP, and new theory, methods, and computational tools are shared with members of the MCS SIG, and then with the broader ISoP community.

Statistics and pharmacometrics (SxP) SIG

The SxP SIG was established to facilitate collaboration between statistics and pharmacometrics to support development of new and innovative MIDD approaches. Although much of the statistical theory and methodology used in pharmacometric models has been developed by statisticians, interaction between pharmacometricians and statisticians has historically been limited. With inferences now being made through the lens of the MIDD framework, as well as the foundation of new regulatory pathways, it is essential that statisticians and pharmacometricians are working together and recognizing our commonalities to improve drug development. SxP is jointly chartered by the American Statistical Association (ASA), with collaborations with the Biopharmaceutical Section of the ASA.

Clinical pharmacometrics (clin PMx) SIG

The mission of the Clin PMx SIG is to promote the application of pharmacometrics to direct patient care. The Clin

PMx SIG aims to (1) create and maintain a forum for communication and collaboration between pharmacometricians with clinical interest and clinicians with an interest in pharmacometrics; and (2) advance the field of personalized medicine through expanded use of pharmacometrics in clinical practice. PMx models have a role in precision dosing to ensure patients can get the optimal dose in the clinic based on their condition and to ensure patients continue to have the optimal dose to maximize efficacy and minimize adverse events after the approval of a new medicine. The Clin PMx SIG is run jointly by ISoP and American College of Clinical Pharmacology (ACCP) and dedicated to fostering an international special interest group of scientists including clinical pharmacologists, pharmacometricians, and clinicians who span a wide range of therapeutic specialties. The Clin PMx SIG is devoted to establishing a platform to support science application from bench to bedside based on multidisciplinary knowledge from both ISoP and ACCP members.

Pharmacometrics data programming (PMxP) SIG

PMxP is the most recent SIG, officially founded in March 2024. Previously, it operated as a working group within SxP SIG from 2016 to 2023. The PMxP SIG aims to create a community for pharmacometric programmers across industry and academia, sharing best practices, challenges, and examples for programmers to support pharmacometrics, QSP, and NCA analysis and reporting. By bringing together a diverse range of expertise, the SIG will address tough challenges, drive innovation and advance the field to new heights.

Integration of PMx approaches to MIDD: across SIG perspectives

The integration of PMx approaches into MIDD decision making is a crucial commonality across the various SIG perspectives. Understanding the vast potential of modeling to answer key questions at all stages of development and knowing what the appropriate model is for different stages and situations is a key skill for pharmacometricians. Selecting the best model begins with understanding the question you need to answer and the data available to determine which model class is most appropriate. In some cases, a simple model (i.e., logistic regression) may be all that is needed to sufficiently answer the question. In other cases, a much more complex model may be required to determine specific dose(s) and regimen(s) that should be studied.

Each SIG has a different focus and modeling approach when addressing dose selection questions. At the American Conference on Pharmacometrics (ACoP14) in 2023, the

SIGs all collaborated together and brought the inaugural cross-SIG special session to briefly introduce the different modeling perspectives of each SIG with examples.

QSP SIG

QSP models and their applications synergistically interact with and inform the work conducted with clinical pharmacometrics (PMx) in accordance with the learn and confirm paradigm of model-informed drug development (MIDD). The integration of QSP and PMx in a modeling framework comes in three flavors, namely (1) parallel synchronization (independent efforts serve as cross-validation), (2) cross-informative use (one approach helps the other), and (3) sequential integration (where one approach precedes the other, creating a framework that can inform decisions along the whole continuum of research and development, including *de novo* therapies with limited data) [4]. In addition, an increasing number of examples are found in the literature (e.g., the cross-validation efforts for dose selection of pembrolizumab [5, 6]; the use of a large QSP model of the cardio-renal system to explain the “unexpected” cardio-protective effect of SGLT2 inhibitors in patients with heart failure [7]).

In addition to a mechanistic, mathematical model, QSP workflows may include the creation of virtual patients (VPs), each one of which is a credible observation in the parameter space, consistent with observed pathophysiology and response to different therapies [8]. As another example of sequential integration, the tumor dynamics were simulated in VPs receiving novel immunotherapies and predict overall survival for a select cohort of VPs using joint tumor growth dynamics – overall survival models utilized in clinical PMx [9].

MCS SIG

Traditional methods for determining drug regimens have included scaling from preclinical data or running lengthy clinical studies [10–14]. Mechanistic modeling has been used to speed and improve dose and regimen selection, as highlighted in multiple overviews [15–20]. In rare and/or pediatric diseases with limited clinical data, mechanistic modeling has been particularly important in determining dosing, and has even enabled regulatory approvals [21]. Advanced modeling methods, including optimal and robust control, are used in other real-world settings, and have the potential to find drug regimens that substantially improve patient outcomes [22–25].

Three examples were presented to illustrate how mathematical and computational approaches are central to MIDD. First, the mathematical evaluation required when

a pharmacometrician sets up a new translational PK-PD model was shown for a cell cycle model and exists in the overlap between MCS and QSP [26]. Second, the power of automated model development was clarified in the context of the impact of sample design on correct model identification [27]. Limited sampling increases the error rate, a conclusion in line with the overlap between MCS and SxP. A third illustration was the derivation of a large-time algebraic relationship between clearance of biological drugs and FcRN-specific parameters by means of asymptotic analysis, in the interface of MCS and QSP. It is envisioned that enhanced understanding of the relationship will contribute to drug design and education alike [28].

SxP SIG

Population pharmacokinetic, pharmacodynamic, and exposure-response analyses are well-established methods that are routinely used for describing variability between subjects, justifying safe and effective doses, and designing trials in populations of interest, at various phases of clinical development. These population analyses are applications of statistical approaches, such as longitudinal mixed-effects modeling, generalized linear models, and time-to-event analyses. The models are largely informed by clinical data and can be as simple as logistic regression or can incorporate complex mechanistic structures.

An example that showcases the collaboration between statistics and pharmacometrics was the development of a disease progression model for systemic lupus erythematosus (SLE) using latent variable modeling [29]. The analysis utilized data from the TransCelerate BioPharma’s Historical Trial Data Sharing Initiative. Programming and biostatistics supported endpoint derivation rules, whereas clinical development, bioinformatics, and preclinical pharmacology provided input on disease etiology and target biology. Health economics and outcomes research contributed important endpoints for the competitive landscape. Advanced analytics, including latent variable modeling and Bayesian statistics, were employed, along with artificial intelligence/machine learning for covariate screening. The disease progression modeling using latent variable framework was also used for model-based meta-analysis [30]. This collaboration enabled a better understanding of the disease, target biology, appropriate modeling framework, and advanced analytics, ultimately improving drug development by incorporating internal and external datasets.

Clin PMx SIG

The primary objective of clinical pharmacometrics is to facilitate the utilization of pharmacometrics in guiding

and improving the provision of direct patient care, thereby enhancing treatment outcomes and optimizing medication regimens. An example of how clinicians utilized Bayesian dosing for precision vancomycin dosing, for a “bench to bedside” approach was presented. Experiential points of implementation were discussed including the practicality of integrating software programs into the electronic medical record for ease of use, clinician training for software use and model selection, and standardization of processes. Patient safety outcomes were assessed pre- and post-implementation: decreased incidence of acute kidney injury, significantly fewer therapeutic drug monitoring laboratory levels obtained, and significantly lower vancomycin trough values were found, demonstrating overall improvement in quality and safety of patient care post-implementation [31]. Practical advice was provided for clinicians to use Bayesian dosing software as a tool to improve patient care. Clinical intuition is still needed, but when Bayesian software (and pharmacometricians) is/are utilized, patients get to their goals faster and safer.

PMxP SIG

The availability of high-quality clinical data is fundamental for pharmacometric analyses. Early collaboration between the pharmacometrician, statisticians, statistical programmers, and the study team is essential for planning what data need to be collected, when, and in which format. The programmers play a key role in developing analysis datasets, identifying appropriate source data to be used for programming the analysis dataset, such as demographics, dosing, lab measurements, pharmacokinetic and pharmacodynamic data. The PMxP SIG recently played a significant role in the design and publication of the new CDISC standard for population PK analysis data. This new standard will help to ensure higher quality data and alignment across different stakeholders.

Discussion

The presentations in the cross-SIG special session at the ACoP 2023 highlighted the many ways in which models are used to help inform drug development decisions and how different groups contributed or collaborated, either directly or indirectly towards these decisions. Whether the goal was to select appropriate doses for a clinical study, translate pre-clinical information to patients, or answer specific clinical questions, the models were able to incorporate available information to make the most informed decision. While these were just a few examples, they nicely illustrated the

importance of collaboration in drug development and the benefits to patients from MIDD approaches.

The interdisciplinary nature of MIDD requires pharmacometricians to understand the contributions of every stakeholder. It is through these collaborations that we are improving drug development and bringing medicines to patients faster. These collaborations are not just across disciplines, but also need to be within the ISoP community and specifically amongst the SIGs (Fig. 2). Since many of the objectives and interests for each SIG overlap with other SIGs (e.g. QSP and SxP are both studying model-based meta-analysis (MBMA), and all of the SIGs are studying artificial intelligence/machine learning (AI/ML) methodology), it is essential that our efforts are more collaborative to take advantage of each other's strengths. The session at ACoP 2023 was the just first step and future sessions will address new opportunities.

The potential of AI/ML methodologies to transform drug discovery and development is at the forefront of our entire industry. As we attempt to identify how and where we can take advantage of opportunities from AI/ML, it is imperative that we work together and exchange knowledge. Decades of prior research from the computer science, physics, and statistical communities has built the foundation we continue developing today. Methods such as random forests, gradient boosting, bagging, lasso, and neural networks, just to name a few, were all originally developed in the 20th century. With each SIG interested in how they can implement AI/ML methods, it is important that we keep in perspective our overlapping interests and look for opportunities to collaborate with each other, especially where we find ourselves doing similar work. Some of these opportunities include data generation, covariate selection approaches, model evaluation, package development, limitations of AI/ML, and identifying ethical boundaries. These are all areas where collaboration will take us further, faster, and safer.

What should the “pharmacometrician” of the future be and how can SIGs contribute?

The SIGs have a collaborative role to play in developing the next generation of pharmacometricians. Training does not end with a diploma, and in this ever-changing MIDD environment, pharmacometricians need to evolve and adapt. They cannot be experts in everything but need a solid “technical” basis as practitioners and need to demonstrate flexibility to move across a variety of (computational) concepts and tools (and application domains). This begs the question, what opportunities are available for technical and professional development. The SIGs need to collaboratively work towards: a) promoting “complex system dynamics thinking” since the initial boundaries of “traditional pharmacokinetics/

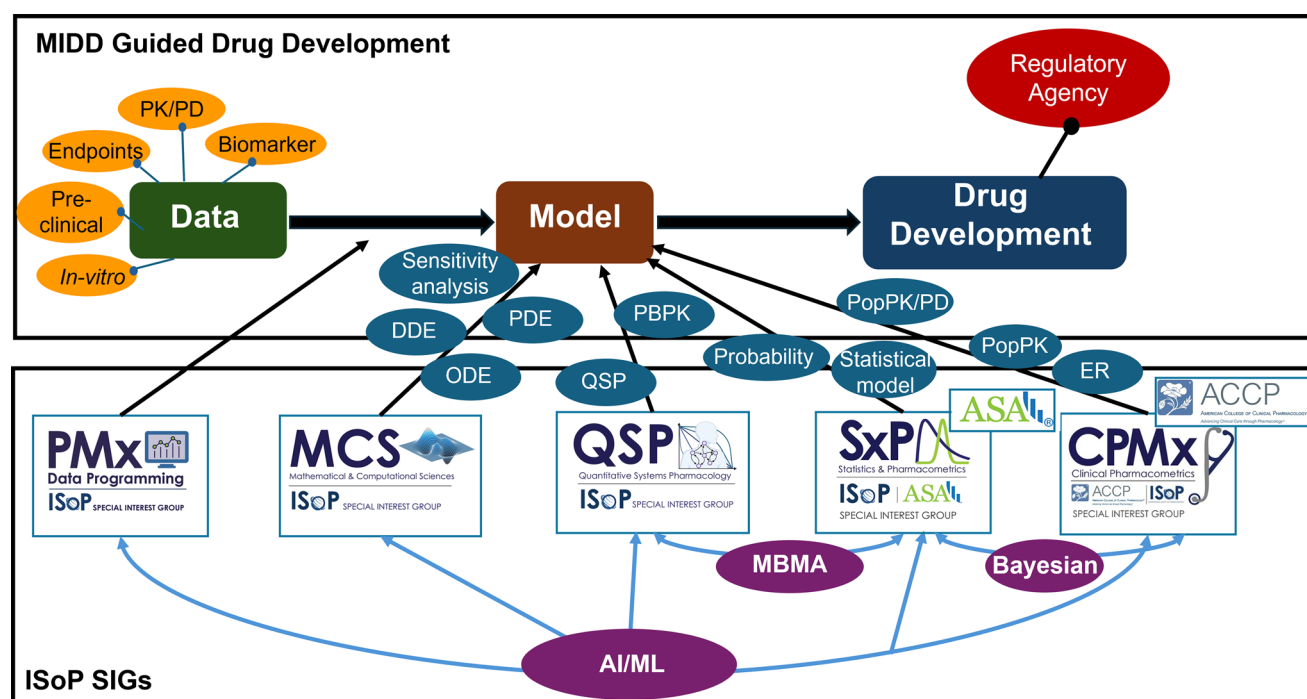


Fig. 2 Cross-SIG networks. An illustrative network diagram that showcases the collaboration across ISoP SIGs. MIDD, model-informed drug development; QSP, quantitative system pharmacology; SxP, statistics and pharmacometrics; CPMx, clinical pharmacometrics; PMx, pharmacometrics; MCS, mathematical and computational sciences; ISoP, International Society of Pharmacometrics; SIG, special interest group; ASA, American Statistical Association; ACCP, American Col-

lege of Clinical Pharmacology; MBMA, model-based meta-analysis; AI/ML, artificial intelligence/machine learning; DDE, delay differential equation; PDE, partial differential equation; ODE, ordinal differential equation; PK/PD, pharmacokinetics/pharmacodynamics; PBPB, physiologically based pharmacokinetics; DDI, drug-drug interactions; ER, exposure-response

pharmacodynamics (PK/PD)” have long been crossed, thus modeling/MIDD needs to cross several scales (basic science to regulatory and business decisions); b) enabling contextualization of diverse data modalities using mathematical and computational representations; c) promoting a culture encouraging cross-SIG interactions and enhancing the ability to work in teams (hence the age-old discipline-specific paradigm needs to be relaxed); d) developing consistent and systematic continuous learning opportunities to engage the workforce at various stages of career development; and last but not least e) emphasizing the need to communicate science.

As the famous observation, commonly attributed to be an African proverb, succinctly says: “If you want to go fast, go alone. If you want to go far, go together”. The SIGs have their own working groups that focus on methodological development, regulatory development, educational opportunities, or other important topics. We believe each SIG should also be looking for opportunities to collaborate with the other SIGs to leverage the different backgrounds of the members. Could our work benefit from knowing about the methods that other SIGs specialize in? Are we duplicating efforts that collaboration with other SIGs would have avoided? Could we provide training and leadership opportunities across

all of the SIGs, to bring forth future industry leaders who understand quantitative modeling? We believe that collaboration across the SIGs will result in better science, better leaders, and better medicines. We look forward to what we will achieve together!

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests Luke Fostvedt is an employee of Pfizer and owns stock in Pfizer. Jiawei Zhou is an employee of Pfizer and owns stock in Pfizer. Anna Kondic is an employee of BMS and owns stock in BMS.

Meghan Pryor is an employee of Johnson & Johnson and owns stock in J&J. Luning Zhuang is an employee of BMS and owns stock in BMS. Jeroen Elassaiss-Schaap is the owner of PD-value B.V. Phyllis Chan is an employee of Genentech and owns stock in Roche. Kosalaram Goteti is an employee of EMD Serono. Neelima Thanneer is an employee of BMS. Jing Su is an employee of Merck & Co and owns stock in Merck. Sihem Ait-Oudhia is an employee of Merck & Co and owns stock in Merck.

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